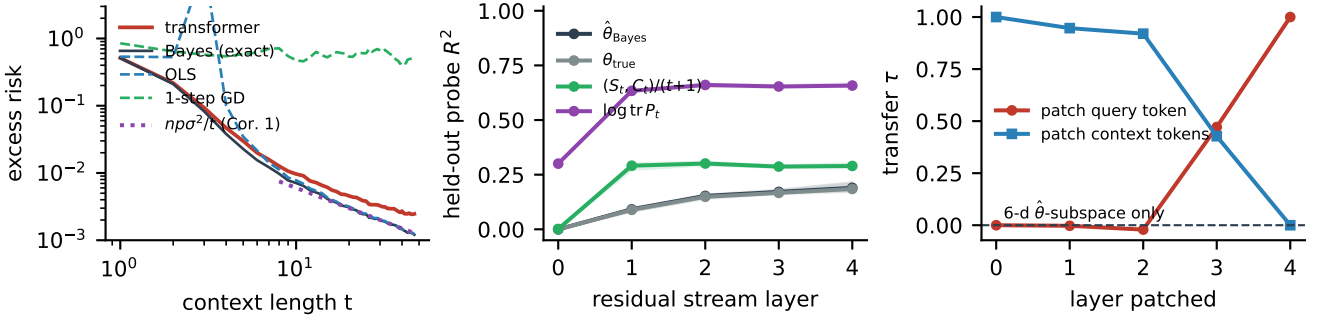
(a) in-context prediction
excess risk
context length t
transformer
Bayes (exact)
OLS
1-step GD
npσ²/t (Cor. 1)
(b) linear probes
held-out probe R²
residual stream layer
θ̂Bayes
θtrue
(St, Ct)/(t+1)
log tr Pt
(c) activation patching
transfer τ
layer patched
patch query token
patch context tokens
6-d θ̂-subspace only